Introduction.

The computer has become an essential part of our everyday lives, playing a vital role in many areas. Today, computers are used widely in fields such as education, where they help students learn and teachers teach more effectively; in medicine, where they assist in diagnosis, research, and treatment; in communication, enabling people around the world to connect instantly; in business, helping companies manage their operations and data; and in entertainment, providing countless hours of games, movies, music, and creative activities. However, computers were not always the fast, small, and powerful machines we are familiar with today. In the early days, computers were massive devices that occupied entire rooms and could only perform basic calculations. The history of computing is a fascinating journey of human innovation, showing how people have always sought to create tools to calculate numbers and process information more quickly and accurately. Over the years, technological advances led to the invention of different types of computers, each designed based on specific factors such as size, purpose, speed, and processing power. This evolution from large mechanical machines to the sleek laptops and smartphones we use today highlights the remarkable progress in computer technology.

1. History of Computing

The history of computing can be divided into different stages:

1.1 Early Calculating Devices

Before modern computers, humans used simple tools to help with calculations:

- **Abacus** (around 3000 BC):
  - Invented by the Chinese.
  - Used for addition, subtraction, multiplication, and division.
  - It is a frame with beads sliding on rods.
- **Napier's Bones** (1617):
  - Invented by John Napier.
  - Used for multiplication and division with rods marked with numbers.
- **Slide Rule** (1632):
  - Developed based on Napier's ideas.
  - Used for multiplication, division, and finding square roots.
- **Pascaline** (1642):
  - Invented by Blaise Pascal.

    - A mechanical calculator that could add and subtract.

- **Leibniz Calculator** (1671):
    - Invented by Gottfried Wilhelm Leibniz.
    - Improved Pascal's design to also do multiplication and division.

1.2 The Beginning of Modern Computing

- **Charles Babbage's Analytical Engine** (1837):
    - Known as the "Father of the Computer."
    - Designed a machine that could perform any calculation.
    - It had a memory (store) and processor (mill), much like modern computers.
    - However, it was never completed in his lifetime.
- **Ada Lovelace**:
    - Worked with Babbage.
    - Known as the "First Computer Programmer" because she wrote the first algorithm for his machine.

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- **Herman Hollerith's Tabulating Machine** (1890):
    - Used punched cards to store data.
    - Helped to process data for the U.S. Census faster.
    - Later became part of IBM (International Business Machines).

1.3 The First Electronic Computers.

- **ENIAC (Electronic Numerical Integrator and Computer)** (1946):
    - Built by John Mauchly and J. Presper Eckert.
    - The first general-purpose, fully electronic digital computer.
    - Very large (filled an entire room).
- **EDVAC (Electronic Discrete Variable Automatic Computer)**:
    - Introduced the idea of storing programs inside the computer (stored program concept).
- **UNIVAC I (Universal Automatic Computer)** (1951):
    - The first commercial computer used for business and government applications.

History of Computers part 1 BBC Documentary

https://www.youtube.com/watch?v=6dME3wgaQpM&list=PL8al-RMvV16iPwtgwp45163YKUmFSAvy1&index=6

https://www.youtube.com/watch?v=zPhZjOAMccU&list=PL8al-RMvV16iPwtgwp45163YKUmFSAvy1&index=7

https://www.youtube.com/watch?v=8NEMUmOwHzU&list=PL8al-RMvV16iPwtgwp45163YKUmFSAvy1&index=8

https://www.youtube.com/watch?v=d4LVp2NiXIY&list=PL8al-RMvV16iPwtgwp45163YKUmFSAvy1&index=9

https://www.youtube.com/watch?v=P75baJbHloI&list=PL8al-RMvV16iPwtgwp45163YKUmFSAvy1&index=10

https://www.youtube.com/watch?v=P75baJbHloI&list=PL8al-RMvV16iPwtgwp45163YKUmFSAvy1&index=10

https://www.youtube.com/watch?v=zKWQAXmpohY&list=PL8al-RMvV16iPwtgwp45163YKUmFSAvy1&index=11

https://www.youtube.com/watch?v=jhvJtJg5g1s&list=PL8al-RMvV16iPwtgwp45163YKUmFSAvy1&index=12

https://www.youtube.com/watch?v=471wnn1y20A&list=PL8al-RMvV16iPwtgwp45163YKUmFSAvy1&index=14

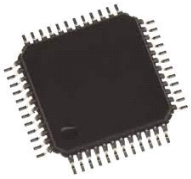
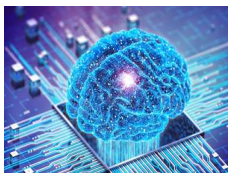
https://www.youtube.com/watch?v=ktZK7Nwpsj4&list=PL8al-RMvV16iPwtgwp45163YKUmFSAvy1&index=15

A Very Brief History of Computing, 1948-2015 - Professor Martyn Thomas CBE

https://www.youtube.com/watch?v=9v7qDKom9nU

1.4 Generations of Computers.

The history of computers is often described in **five generations**:

Generation	Technology Used	Key Features
First (1940–1956)	Vacuum tubes	Large size, lots of heat, slow
Second (1956–1963)	Transistors	Smaller, faster, more reliable
Third (1964–1971)	Integrated Circuits (ICs)	More compact, more powerful
Fourth (1971–Present)	Microprocessor 	Personal computers (PCs), mobile devices
Fifth (Present and Beyond)	Artificial Intelligence (AI) 	Smart computers that can learn and make decisions

2. Various Kinds of Computers.

Computers can be categorized based on their size, purpose, and performance:

2.1 Based on Purpose.

- **General Purpose Computers**:
  - Used for many different tasks (e.g., laptops, desktops).
  - Can run various programs like Word processors, games, educational software.
- **Special Purpose Computers**:

- Designed for a specific task.
- Examples: ATM machines, traffic control systems, medical equipment.

2.2 Based on Size and Power.

- **Supercomputers**:
  - Most powerful and expensive.
  - Used for weather forecasting, scientific research, and simulations.
  - Example: IBM Summit, Fugaku (Japan).
- **Mainframe Computers**:
  - Large computers used by big organizations for bulk data processing.
  - Example: Banks, insurance companies.
- **Minicomputers** (also called Mid-range Computers):
  - Smaller than mainframes but still powerful.
  - Used by medium-sized businesses.
- **Microcomputers**:
  - Smallest and most common type.
  - Examples: Personal computers (PCs), laptops, tablets, smartphones.

2.3 Other Types

- **Workstations**:
  - High-performance single-user computers for technical or scientific applications.
  - Used by graphic designers, engineers, architects.
- **Servers**:
  - Provide services and manage network resources.
  - Example: Web servers, file servers.
- **Embedded Computers**:
  - Built inside other machines to control functions.
  - Example: Washing machines, microwaves, cars.

The journey of computing started with simple counting tools and led to the invention of highly advanced machines. Understanding the history and types of computers helps us appreciate how human needs and

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creativity have shaped technology. Today, computers are essential tools that make our lives easier, faster, and more connected.

Review Questions

1. Fill in the blanks:

- a) _____ is known as the Father of the Computer.
- b) The first electronic general-purpose digital computer was _____.
- c) Computers used in mobile phones and laptops belong to the _____ generation.

2. Short Answer Questions:

- a) What is the difference between a general-purpose and a special-purpose computer?
- b) Name two early calculating devices before the invention of electronic computers.
- c) Give two examples of embedded computers.

3. Long Answer Question:

- a) Briefly explain the five generations of computers with their key features.

Reference Book.

Compiled by J.K.S.Pathirana (Dr. Prof.Study., MA in TE, BEd, Hig Dip in ICT, Dip in DBA) for CINEC BEd (Hon) Top up Degree.

- Rajaraman, V. (2018). *Fundamentals of Computers* (6th ed.). Prentice Hall of India.
- Sinha, P.K. & Sinha, Priti. (2020). *Computer Fundamentals*. BPB Publications.

Unit 2: Understanding Computers — Systems, Functional View, Hardware, Peripherals, and Communication Equipment.

1. Basic Concept of a System

A **system** is a set of **interrelated components** working together towards a **common goal** by accepting inputs, processing them, and producing outputs.

Definition:

A system is an organized collection of parts (subsystems) that are highly integrated to accomplish an overall objective.

Characteristics of a System:

- **Input**: Receives information or materials.
- **Process**: Transforms the input into useful output.
- **Output**: The final result of the processed input.

- **Feedback**: Information about output that is used to make corrections.
- **Control**: Guides the system to function properly.

Example:

- A school is a system:
  - Input: Students, teachers, curriculum.
  - Process: Teaching and learning activities.
  - Output: Educated students.

In Computing:

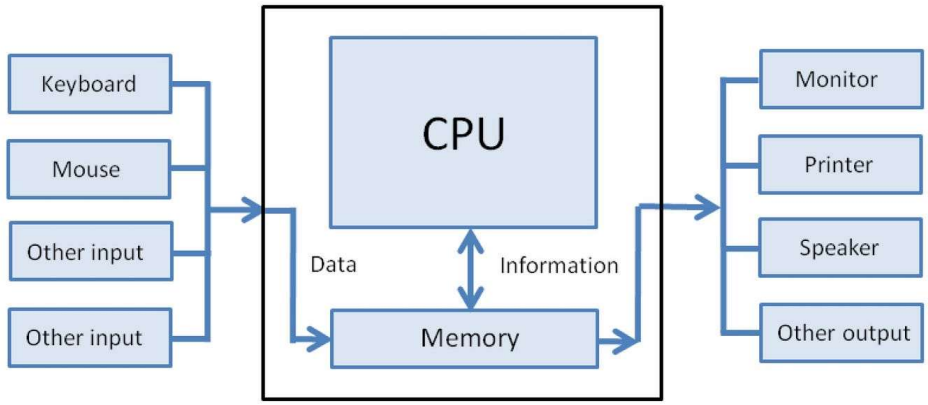
A **computer system** consists of hardware, software, data, and users, all working together to perform useful tasks.

2. Functional View of a Computer

When we look at a computer from a **functional perspective**, it performs **five major operations**:

Operation	Purpose
Input	Accepts data from the user (e.g., keyboard, mouse).
Storage	Saves data temporarily (RAM) or permanently (hard drive).
Processing	Performs operations on the data using CPU.
Output	Produces results (e.g., monitor, printer).
Control	Manages and directs the way the above components interact using the control unit inside the CPU.

Simple Diagram:



Example:

Typing a document:

- Input: Keyboard typing.
- Processing: CPU processes keystrokes.
- Storage: File saved on hard drive.
- Output: Document displayed on the screen.

3. Hardware

Hardware refers to the **physical components** of a computer that you can touch and see.

Main Categories of Hardware:

- **Input Devices**: Bring data into the computer.
- **Processing Devices**: Perform computations (e.g., CPU).
- **Storage Devices**: Save data.
- **Output Devices**: Display or produce the results.

Central Processing Unit (CPU)

- Known as the "brain" of the computer.
- It consists of:

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  - **Arithmetic and Logic Unit (ALU)**: Performs calculations and logical comparisons.
  - **Control Unit (CU)**: Directs the flow of operations.
  - **Registers**: Small, fast storage locations for immediate data processing.

Types of Memory:

- **Primary Memory**:
  - RAM (Random Access Memory) — temporary, volatile.
  - ROM (Read Only Memory) — permanent, non-volatile.
- **Secondary Memory**:
  - Hard Disk Drives (HDD), Solid State Drives (SSD).

4. Hardware Peripherals

Peripheral Devices are devices connected to the computer externally to add functions.

Types of Peripherals:

Input Devices

- **Keyboard**: For text input.
- **Mouse**: For pointer-based control.
- **Scanner**: Converts physical documents into digital form.
- **Microphone**: For voice input.
- **Webcam**: For capturing video.

Output Devices

- **Monitor**: Visual output.
- **Printer**: Physical (hard copy) output.
- **Speakers**: Audio output.
- **Projectors**: Enlarges the computer display onto a larger screen.

Input/Output (I/O) Devices

- **Touchscreen**: Both input (touch) and output (display).
- **External Hard Drives**: Both reading (input) and writing (output) data.

Ports and Connectors:

- **USB (Universal Serial Bus)**: Connects peripherals.
- **HDMI (High Definition Multimedia Interface)**: For high-quality video and audio output.

5. Communication Equipment and Concepts.

In today's interconnected world, computers often communicate with each other through networks. **Communication equipment** refers to devices that enable this sharing of data.

Key Communication Equipment:

- **Modem**:
    - Converts digital signals into analog signals (and vice versa).
    - Used for internet connections over telephone lines.
- **Router**:
    - Directs data traffic between computers and networks.
    - Provides wireless internet in homes and offices (Wi-Fi).
- **Switch**:
    - Connects multiple devices in a network.

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    - Sends data directly to the intended device.
- **Hub**:
    - A basic networking device that copies data to all devices.
- **Network Interface Card (NIC)**:
    - Installed inside computers to connect to networks.
- **Access Points**:
    - Allow wireless devices to connect to a wired network.

Basic Communication Concepts

- **Network**: A group of computers connected together to share resources.
- **LAN (Local Area Network)**: Small area like a school or office.
- **WAN (Wide Area Network)**: Large area connection (e.g., the Internet).
- **Bandwidth**: The amount of data transmitted in a fixed amount of time.
- **IP Address**: A unique number assigned to each device in a network.
- **Protocol**: Set of rules for data communication (e.g., TCP/IP).

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A computer is not just a machine — it is a **system** made up of different **hardware components**, **peripheral devices**, and **communication tools** working together to perform useful tasks. Understanding these basic concepts is essential for students of IT, as it builds the foundation for studying more advanced topics like networking, system design, and troubleshooting.

Review Questions.

1. Fill in the blanks:

- a) A _____ is the brain of the computer where processing takes place.
- b) Devices like a mouse and keyboard are called _____ devices.
- c) A _____ connects computers and networks and provides wireless internet.

2. Short Answer Questions:

- a) What is a system? Give one real-life example other than a computer.
- b) List any three input devices and three output devices.
- c) What is the function of a modem in a network?

3. Long Answer Question:

- a) Explain the functional view of a computer with a neat diagram.
- b) Describe different types of hardware peripherals with examples.
- c) What is communication equipment? List and explain any three such devices.

Reference Book

- Sinha, P.K. & Sinha, Priti. (2020). *Computer Fundamentals*. BPB Publications.
- Leon, Alexis & Leon, Mathews. (2017). *Introduction to Computers*. Vikas Publishing House.

Unit 3: Computer System Architecture, Memory, Software, and Operating Systems

1. Computer System Architecture

Computer System Architecture refers to the **design and structure** of a computer system — how its components interact and function together.

Main Components of Computer Architecture:

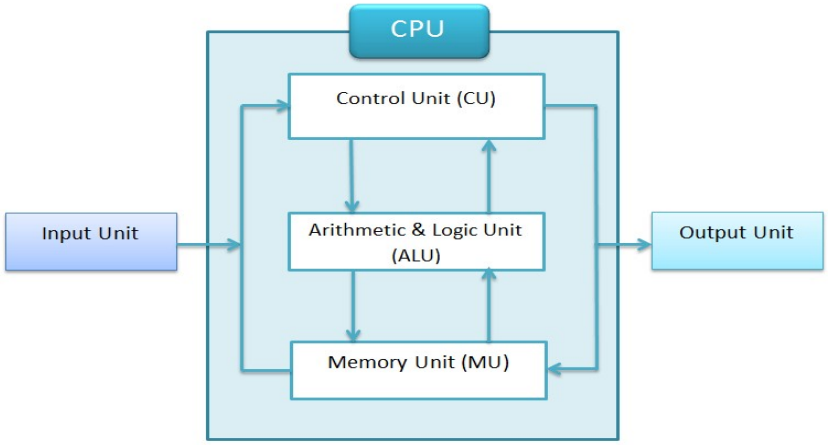
Component	Description
Input Unit	Devices that input data into the computer (keyboard, mouse).
Output Unit	Devices that present information to users (monitor, printer).
Memory Unit	Storage for data and instructions.
Control Unit (CU)	Directs operations and controls signals between units.
Arithmetic Logic Unit (ALU)	Performs calculations and logical comparisons.
Central Processing Unit (CPU)	Combination of ALU + CU; brain of the computer.
Secondary Storage	Long-term data storage (hard drives, SSDs).

Von Neumann Architecture.

Most modern computers follow this model:

- Single memory space for instructions and data.
- Instructions executed sequentially.
- Control unit fetches instructions one by one, decodes them, and executes them.

Diagram (Simplified):



2. Computer Memory

Memory is used to store data, instructions, and results, either temporarily or permanently.

Types of Memory:

Type	Characteristics	Examples
Primary Memory	Directly accessible by CPU, volatile.	RAM, ROM
Secondary Memory	Not directly accessible by CPU, non-volatile.	Hard Disk, SSD
Cache Memory	Very fast, between CPU and RAM, stores frequently used data.	L1, L2, L3 Cache
Virtual Memory	Extension of RAM on the hard disk, slower.	Pagefile on Disk

3. Components and Organization of Computer Memory

Primary Components:

- **RAM (Random Access Memory):**
  - Temporary storage.
  - Data is lost when power is off.
- **ROM (Read Only Memory):**
  - Permanent storage.
  - Contains BIOS (Basic Input Output System) needed for booting.

Memory Hierarchy:

- Faster but smaller memories are closer to CPU.
- Slower but larger memories are farther.

Level	Type	Speed	Size
1	CPU Registers	Fastest	Smallest
2	Cache Memory	Very Fast	Small
3	Main Memory (RAM)	Fast	Medium
4	Secondary Storage	Slow	Very Large

4. Application Software

Application Software is software designed to help the user perform specific tasks.

Categories:

- **Productivity Software**: Word processors (MS Word), spreadsheets (Excel), presentation software (PowerPoint).
- **Media Software**: Video editors, image editors (Photoshop).
- **Educational Software**: Language learning apps, simulation programs.
- **Business Software**: Accounting software (Tally), Customer Relationship Management (CRM) systems.

Examples:

- Microsoft Office Suite
- Google Chrome
- Adobe Photoshop

5. System Software

System Software manages and supports the computer's basic functions, acting as a bridge between hardware and user applications.

Types of System Software:

- **Operating System (OS)**: Manages hardware and software resources.
- **Utility Programs**: Perform maintenance tasks (disk cleanup, antivirus).
- **Device Drivers**: Enable the OS to control hardware devices.

- **Language Translators**: Convert programming languages into machine language (compilers, interpreters).

Examples:

- Windows 10, Ubuntu (Operating Systems)
- WinRAR (Utility)
- Printer Drivers (Device Driver)
- GCC (Compiler for C programming)

6. Functionality of Operating Systems

An **Operating System (OS)** is essential software that controls the computer hardware and provides services to applications and users.

Main Functions:

Function	Description
Process Management	Handles creation, scheduling, and termination of processes.
Memory Management	Allocates and deallocates memory to applications.
File System Management	Manages reading/writing of data to storage devices.
Device Management	Controls and manages hardware devices.
Security and Access Control	Protects data and resources from unauthorized access.
User Interface	Provides graphical or command-line interaction for users.

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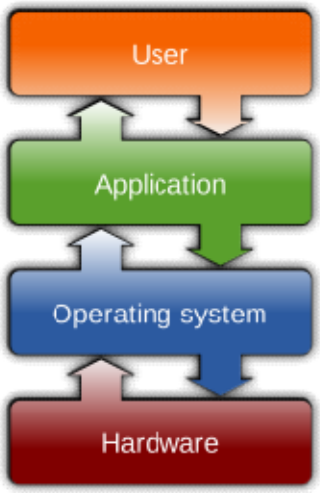
Examples of Operating Systems:

- **Desktop OS**: Windows, MacOS, Linux
- **Mobile OS**: Android, iOS

Simplified View of OS Responsibilities:



Understanding **computer system architecture**, **memory organization**, and **software types** is crucial for IT students. These concepts form the foundation of computer operation, performance, and user interaction. The **Operating System** plays a central role in ensuring all hardware and software components work together smoothly.

Review Questions

1. Fill in the blanks:

- a) The CPU consists of the Control Unit and the _____.
- b) _____ memory is volatile and loses data when the power is off.
- c) An _____ system acts as a bridge between the user and the hardware.

2. Short Answer Questions:

- a) What is the Von Neumann Architecture?
- b) Name three types of system software.
- c) Differentiate between primary memory and secondary memory.

3. Long Answer Questions:

- a) Explain the memory hierarchy with examples.
- b) Describe the major functions of an Operating System.
- c) What are application software? Give examples of different categories.

Reference Book

- Sinha, P.K., & Sinha, Priti. (2020). *Computer Fundamentals*. BPB Publications.
- Tanenbaum, Andrew S. (2015). *Structured Computer Organization*. Pearson.

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Assignment: ICT and Its Role in the Field of Education

Assignment Title:

"The Evolution of Computing and Its Impact on Education"

Assignment Purpose:

This assignment will help students demonstrate their understanding of the **foundations of ICT** and **analyze how ICT shapes the modern educational landscape**, aligned with the module's intended learning outcomes (ILOs).

Assignment Instructions:

Prepare a **structured report** (approximately **2000–2500 words**) covering the following tasks. Use **headings, subheadings, diagrams, and real-world examples** from the field of education wherever possible.
Include references where needed.

Assignment Tasks:

Task 1: Evolution of Computers

Compiled by J.K.S.Ruthirangi (Dr. Prof.Study, MA in TE, BEd, HigDip in ICT, Dip in DBA) for CINEC BEd (Hons) Top up Degree.

- Describe the **evolution of the computer** from early mechanical devices to modern systems.
- Identify and explain the **five generations** of computers.
- Focus particularly on **how each generation impacted educational practices** (e.g., calculators in classrooms, computer labs, online learning).

Expected Output:
300–400 words + Timeline or simple visual chart (optional bonus)

Task 2: Basic Computer Architecture and Operations

- Define **basic computer architecture**.
- Explain **how computers work**: input, processing, storage, and output operations.
- Create a **simple labeled block diagram**.
- Give an example of how understanding computer architecture helps teachers/students use computers effectively in education.

Expected Output:
300–400 words + Diagram

Task 3: Basic Components of a CPU and Program Execution

- List and explain the **main components of a CPU** (ALU, CU, Registers, Cache).
- Describe **step-by-step how a simple program** (e.g., a math quiz application) is executed inside a CPU.
- Highlight why it is important for educators to have a **basic understanding** of how computers process tasks.

Expected Output:

300–400 words

Task 4: Computing and Education: Mutual Impact

- Analyze **how computing has changed education** (e.g., virtual classrooms, e-learning, assessment tools).
- Explain how **education influences the development of ICT tools** (e.g., the demand for educational apps, LMS systems like Moodle).

Expected Output:

300–400 words

Include examples like smartboards, online exams, etc.

Task 5: IT and Related Disciplines in Education

- Explain the relationship between **Information Technology** and other fields such as **educational psychology, curriculum design, instructional technology**.
- Use examples showing how **knowledge from other disciplines informs ICT practices in education** (e.g., using cognitive load theory when designing e-learning platforms).

Expected Output:

300–400 words

Task 6: Changes in Application Domains Due to IT in Education

- Explain **how and to what extent ICT has transformed different areas of education**, such as:
    - Content delivery (e-learning, MOOCs)
    - Student assessment (online quizzes, automated feedback)
    - Administration (student information systems)
- Discuss **both positive changes and challenges**.

Expected Output:

300–400 words

Submission Requirements:

- Format: Typed, Times New Roman 12 pt font, 1.5 line spacing.
- File format: PDF or Word document.
- Include a **cover page** (title, student name, registration number, date).
- Cite references in **APA format** (minimum 3 sources).

Evaluation Rubric:

Criteria	Marks
Understanding of Concepts	20
Relevance to Education Field	20
Structure and Presentation	15
Use of Examples and Diagrams	15
Language and Grammar	10
References and Citations	10
Creativity and Critical Thinking	10
Total	100

Reminder:

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This is not only a theoretical report — you are expected to **connect all your answers to the field of education** with **real-life examples** (e.g., computers in classrooms, mobile learning apps, etc.).